MLVU information sheet

Group number:
20

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Software used: *pandas, numpy, PyTorch, scikit-learn, imbalanced-learn (SMOTE), matplotlib, seaborn, scipy*

Use of AI tools: *Grammarly and Chat GPT were used to improve the clarity and quality of the written report. Chat GPT was also used minimally to assist in data set balancing. Claude was used to support the understanding of mathematical formulas and evaluation metrics.*

Link to code: `https://github.com/Lucyyru/Diabetes_Type2_Prediction`

A Comparison of Machine Learning Models for Diabetes Prediction

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Abstract. Type II diabetes is a serious condition that affects primarily older adults. In particular, Type II diabetes is strongly associated with lifestyle factors such as smoking, physical activity, diet, and health history. Currently, diagnosis typically requires blood tests, which can be invasive and costly. To address this, the paper investigates how Logistic Regression, Support Vector Machines, and Neural Networks compare in predicting Type II diabetes using lifestyle data. To evaluate this, the dataset was balanced and split into training, validation and test sets. It contains 21 features that capture relevant lifestyle and health indicators. Using this data, Neural-Networks, Support Vector Machines and Logistic Regression were evaluated to determine their effectiveness in diabetes prediction. In this setup, all models were implemented as binary classifiers distinguishing between diabetic and non-diabetic cases. The results indicate that all three models achieved very similar performance across the metrics evaluated, suggesting that further investigation is required to determine whether any model offers a consistent advantage.

Keywords: Diabetes Type II Prediction · Support Vector Machines · Linear Regression · Neural Networks.

1 Introduction

The number of people with diabetes has increased significantly, exceeding 830 million cases worldwide by the end of 2022, with more than 95% classified as Type II diabetes [1]. It is a chronic condition that occurs when the pancreas is unable to produce enough insulin or when the body cannot make effective use of it. Although genetic factor plays an important role, there are other risk factors, such as overweight, lack of exercise, among others, that increase the risk of developing the disease. Early diagnosis is essential to prevent the development of severe complications such as serious damage to nerves and vessels, increasing the risk of heart attacks, stroke, and kidney failure [2]. However, traditional diagnostic methods, such as regular check-ups and blood tests, can be invasive, costly, and resource-intensive [2]. In recent years, machine learning techniques have had a major impact on healthcare, as these models can be trained to learn patterns from existing data. This is particularly valuable given that healthcare sector, in general, has a large number of databases available and can be very useful to develop this kind of techniques [3]. The purpose of this project is to predict whether a patient is likely to have type II diabetes, using an available dataset, extracted from the Centers for Disease Control and Prevention (CDC) databases. To achieve this, we trained and evaluated three supervised classification machine learning models: logistic regression, Support Vector Machines (SVMs) and Neural Networks (NNs). Their performance is compared using multiple evaluation metrics, with particular emphasis on the Receiving Operating Characteristic Area Under the Curve (ROC-AUC). Therefore, this study tackles the following research question: How do logistic regression, Support Vector Machine (Linear SVMs), and Neural Networks differ in the diabetes type 2 prediction?

2 Preparations Before Implementation

Before implementing the models, the dataset was prepared through data preprocessing. This stage involved several steps, including splitting the data into training, validation, and test sets, as well

as addressing class imbalance [4]. In real-world scenarios, the majority of people do not have diabetes, which resulted in a highly imbalanced dataset. To mitigate this, the Synthetic Minority Oversampling Technique (SMOTE) was applied to the training set only. SMOTE is an oversampling method that generates new synthetic examples of the minority class, thus improving class balance [5]. The validation and test sets were left unchanged to ensure a realistic evaluation of the performance of the model. Furthermore, the original data set also contained pre-diabetes, which were not investigated in this study, and therefore were removed during data preprocessing. Furthermore, if a person has diabetes, the model outputs 1, and if a person does not have diabetes, the model outputs 0.

3 Background and Methods

All of the models were implemented using Python. [1]

3.1 Logistic Regression

Logistic regression is a machine learning classification algorithm that uses the logistic sigmoid function to produce values between 0 and 1. Its sigmoid function helps to find the relationship between the independent and outcome variables, assuming a linear decision boundary. This model has become increasingly popular in healthcare and medical research settings, used in various machine learning algorithms that aid in diagnosis [6]. It has also gained popularity in diabetes classification research as one of the most popular classification methods [7]; however, these machine learning methods have not yet been shown to perform better than statistical methods used regularly in medicine [8]. Logistic regression is particularly well-suited in clinical research where the outcome variable is binary [6], making it a natural candidate for the diabetes classification problem, where the target variable shows its presence or absence.

Logistic regression models the conditional probability of a binary outcome. x denotes the feature vector and $y \in \{0, 1\}$ the class label. The model defines the hypothesis function as follows:

$$h_w(x) = P(y = 1 \mid x) = \sigma(w^T x + b)$$

where w is the vector of the model parameters, b is a bias term, and σ is the sigmoid function given by:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The model parameters are chosen by minimizing the negative log likelihood of the data. For a dataset of size n, this corresponds to the following loss function:

$$L = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

This loss function penalizes wrong predictions by assigning a high cost when the predicted probability deviates from the true class. To control model complexity and improve generalization, a regularization term was added to the objective function, resulting in the regularized loss:

$$L = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] + \frac{1}{C} R(w)$$

Where $R(w)$ is a penalty on the magnitude of the parameter vector. The optimal parameters are obtained by solving the following optimization problem:

$$(w^*, b^*) = \arg\min_{w,b} L(w, b)$$

[1] Some implementation were informed by the following YouTube tutorials: YouTube tutorial for Logistic Regression, YouTube tutorial for Neural Networks and YouTube tutorial for Support Vector Machines.

For the logistic regression model, eight features were selected based on distributional separation between classes, assessed via box plots. Those with clear differences between classes were retained, while those showing near identical distributions were excluded, as they provided limited discriminative information for the classifier. Before model training, all features were normalized using min-max scaling:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

The scaling parameters were computed exclusively on the training set and subsequently applied to the validation and test sets. This ensures that no information from unseen data influences the training process.

Hyperparameters were optimized via randomized search over predefined distributions, as it provides a more efficient exploration of the parameter space compared to exhaustive grid search, employing 5-fold cross-validation on the training set. A 10-fold cross-validation procedure was used to estimate the variance of the final model. The final model achieved an AUC of 0.829 ± 0.004, indicating stable performance across different data splits. The validation set was used only for final model evaluation, and the test set was used once to find final results. The regularization parameter was chosen using $\text{LogUniform}(10^{-3}, 10^{3})$, and model configurations were evaluated using 5-fold cross-validation on the training set, and the best-performing configuration was selected. Finally, the hyperparameter tuning returned the following parameters:

Parameter	Optimal C	Max Iterations	Regularization Penalty	Optimization Solver
Value	339.82	500	L1	liblinear

Table 1. Optimal hyperparameters for the logistic regression model

The hyperparameter C governs the regularization strength, where smaller values impose stricter penalties on parameter magnitudes, and larger values, such as the optimal $C = 339.8$ found in the experiments, permit a closer fit to the training data. Two regularization schemes were considered: L1, which induces sparsity by driving select coefficients to zero, and L2, which penalizes large weights without eliminating them. The selection of L1 through hyperparameter tuning implies that several features contributed minimally to the decision boundary and were consequently suppressed, consistent with the earlier observation that many features showed limited distributional separation between classes. Furthermore, a classification threshold of 0.3 was adopted in place of the standard 0.5, as the default threshold is generally unsuitable for imbalanced classification problems [9], and in a medical context, undetected diabetic cases carry a greater clinical cost than false positives.

3.2 Support Vector Machines

Support vector machines (SVMs) are a set of supervised learning methods, used for regression and classification tasks. This type of method can model both linear and non-linear decision boundaries, where non-lineary separable data can be handled by using the kernel trick for deriving non-linear functions and making them linear. However, the regular SVM model is not well known for being implemented in large datasets or for highly unbalanced datasets. If the number of features is much larger than the number of instances, the complexity of the program becomes extremely expensive to compute, then the linear SVM is a better option. The latter is a good option for large, sparse data, as the dataset used consists of 200.000 data and is highly unbalanced [10]. In this work, the main focus is on classification, using a linear Support Vector Machine.

These models work to find a hyperplane called the “decision boundary” that separates the data points from each different class, in this case diabetes or non-diabetes. Since the study focuses on linearly separable data, infinitely many boundary decision hyperplanes may exist [11]. Therefore, this model finds the best fitting decision boundary among the hyperplane by choosing the one with the largest margin, consisting of two parallel lines.

The decision boundary is defined as:

$$w \cdot x + b = 0 \tag{1}$$

with margin boundaries:

$$w \cdot x + b = \pm 1 \tag{2}$$

The margin is defined as:

$$\text{margin} = \frac{2}{\|w\|} \tag{3}$$

Maximizing the margin is equivalent to minimizing:

$$\min_{w,b} \frac{1}{2}\|w\|^2 \quad \text{subject to} \quad (w \cdot x_i + b) \geq 1 \tag{4}$$

Optimization primal problem:

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C\sum_i \xi_i \quad \text{subject to} \quad y_i(w^T x_i + b) \geq 1 - \xi_i \tag{5}$$

The first one belongs to the positive class , and the latter one to the negative (2). The margin is the distance between the decision boundary (1) and the closest data points (3). It is important to look for minimizing the complexity of the model by maximizing the margin between classes (4), by using the optimization formula. Moreover, it was chosen to use the primal optimization form as Chapelle (n.d.) discusses in his paper, both forms, dual and primal, should be equivalent, although primal show to be superior computationally wise when it comes to approximate solutions and minimizing the complexity of the model.

The regularization parameter C (L2), is used to control the softness of the margins. In this model, C was sampled from a logarithmic range $[10^{-3}, 10^2]$, allowing exploration between soft margin errors to harder margin values. When C is large, the penalty is large and the strength of regularization is small, similarly, the other way around. As shown by Mantovani et al. (2015), this strategy has a predictive performance equivalent to other hyperparameter tuning methods such as grid search, but with less computational costs.

Due to computational constraints, a 3-fold-cross validation was chosen. The model was trained and evaluated with ten different parameter combinations, resulting in a total sample of 30 model evaluations. For the loss function, the hinge and squared hinge are being used, which determine how the misclassification penalties are computed.

Parameter	Optimal C	Max Iterations	Regularization Penalty	Optimization Solver
Value	7.742636826811277	10000	L2	liblinear

Table 2. Optimal hyperparameters for the SVM model

3.3 Neural Networks

Neural Networks are machine learning models that consist of layers of interconnected neurons, which learn patterns to map input data to output predictions [12]. In this study, the model implemented is a feedforward neural network for binary classification. A feedforward network is structured such that each node in one layer is connected to every node in the next layer, with each connection assigned a different weight [13]. As a result, the objective of a neural network is to transform input data into meaningful outputs [14]. In particular, feedforward architectures propagate information strictly in a forward direction, from input to output layers. Within this framework, the layers perform the following functions [15]:

- Input layer: Receives the raw input data, which in this case consists of a person's lifestyle information.
- Hidden layers: Transform the input data through weighted connections, where each neuron's activation depends on the input values and associated weights.
- Output layer: Produces the final prediction based on the activations of the previous layer; in this case, the output indicates 1 for "diabetes" or 0 for "non-diabetes".

In terms of architecture, the model consists of 21 input nodes, 128 neurons in the first hidden layer, 64 neurons in the second hidden layer and 1 output node. To train this model, the parameters were optimized using the Adam optimizer with a learning rate of 0.001. The Adam optimizer updates weights using gradients from backpropagation along with adaptive learning rates for each parameter. This optimizer was selected because it adapts learning rates per parameter, resulting in faster convergence. The parameter update rule of the Adam optimizer is given by:

$$v_t = \beta_1 v_{t-1} + (1 - \beta_1) g_t \tag{6}$$

$$s_t = \beta_2 s_{t-1} + (1 - \beta_2) g_t^2 \tag{7}$$

$$\Delta w_t = -\eta \frac{v_t}{\sqrt{s_t} + \epsilon} \tag{8}$$

$$w_{t+1} = w_t + \Delta w_t \tag{9}$$

where η is the learning rate, g_t is the gradient at time t, v_t and s_t are the first and second moment estimates, and β_1, β_2 are the hyperparameters.

Additionally, ReLU activations were applied after each hidden layer to mitigate vanishing gradients and introduce non-linearity, enabling the model to capture complex relationships. The ReLU activation function is given by:

$$f(x) = \max(0, x)$$

where x is the input to the neuron. The model was trained by minimizing the binary cross-entropy loss, which is standard for binary classification. For numerical stability, the sigmoid activation was incorporated directly into the loss function, allowing it to operate on the model's logits. The loss function is defined as:

$$L = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

where N is the number of samples, y_i is the true label for the i-th sample (0 or 1), and p_i is the predicted probability that the i-th sample belongs to the positive class. Training was performed over 100 epochs using forward passes, loss computation, backpropagation, and optimization steps. At the output stage, the final layer produces a real-valued logit z. Formally, the model can be expressed as:

$$h^{(1)} = \mathrm{ReLU}(W_1 x + b_1) \tag{10}$$

$$h^{(2)} = \mathrm{ReLU}(W_2 h^{(1)} + b_2) \tag{11}$$

$$z = W_3 h^{(2)} + b_3 \tag{12}$$

Finally, since the model outputs a binary prediction, the final scalar is converted into a probability with the following sigmoid:

$$\hat{p}(y = 1 \mid x) = \sigma(z) = \frac{1}{1 + e^{-z}}$$

where $\hat{p}(y = 1 \mid x)$ is the predicted probability that the input x belongs to the positive class, $y \in \{0, 1\}$ is the true label, x is the input feature vector, z is the output of the neural network (i.e., $z = W_3 h^{(2)} + b_3$), $\sigma(z)$ is the sigmoid function, and e is Euler's number.

Regarding evaluation, training was performed on the training set, with a validation set used for performance monitoring. The test set was reserved exclusively for the final evaluation of the model. Furthermore, hyperparameters, including the number of hidden layers, neurons per layer, and learning rate, were selected using randomized search, which provides a more efficient exploration of the parameter space than exhaustive grid search. To estimate the variability of model performance and reduce dependence on a single train-test split, 10-fold cross-validation was performed. The final model achieved an AUC of 0.825 ± 0.004, indicating stable performance across different data splits.

4 Results

To address the research question, all models were evaluated using ROC-AUC as the primary metric, due to its robustness to class imbalance and threshold-independent nature [16]. Precision and recall are used as secondary evaluation metrics, as they focus specifically on the positive class and therefore provide a more informative assessment in the presence of class imbalance. In contrast, accuracy is not considered a reliable metric in this context, although it is included in the results table for completeness. Table 3 summarises the performance of all three models, and Figure 1 presents these results visually.

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
SVM	0.7307	0.3159	0.7705	0.4481	0.824
Logistic Regression	0.5931	0.2468	0.9095	0.3882	0.8182
Neural Networks	0.7488	0.3732	0.6345	0.4700	0.8253

Table 3. Performance comparison of machine learning models on the test set

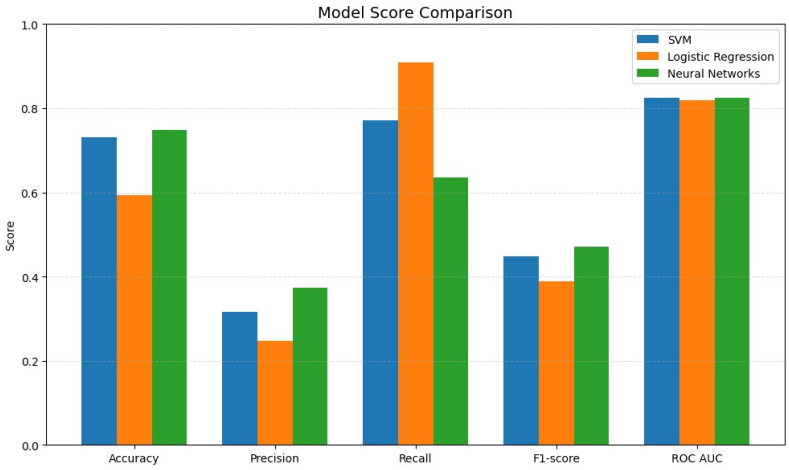

Fig. 1. Comparison of model performance across evaluation metrics

All three models achieved similar ROC-AUC scores of around 0.82, suggesting that these models are similar classifiers. The difference is shown, however, in the other metrics, such as recall and precision. Logistic regression achieves the highest recall (0.91) at the cost of precision (0.25). This is likely due to lowering the classification threshold from the default 0.5 to 0.3, which shifts the prediction towards the positive class. In a medical context, this tradeoff is on purpose, since a false negative carries a higher clinical cost than a false positive. SVM, using the default threshold, achieves the most balanced profile, with the highest F1 score, while Neural Networks falls between

the two. To assess the variance of these results, various cross validations were performed. For logistic regression, a 10-fold cross-validation was performed, which showed a mean AUC of 0.829 with a standard deviation of 0.004, indicating consistent performance with negligible variance. Similarly, a 3-fold cross-validation was performed on the SVM, showing a mean AUC of 0.835 with a standard deviation of 0.001. Finally, Neural Networks, evaluated with 10-fold cross-validation scores, performed poorly compared to the other models, with a mean AUC of 0.817 and a standard deviation of 0.004. All these models score very similarly in AUC, however, SVM scores the strongest with an AUC of 0.835 and a 0.001 standard deviation. While SVM scores best out of the three models, it is worth noting that scores between models vary by approximately 0.02, which makes the differences between them negligible. Figure 2 illustrates these results.

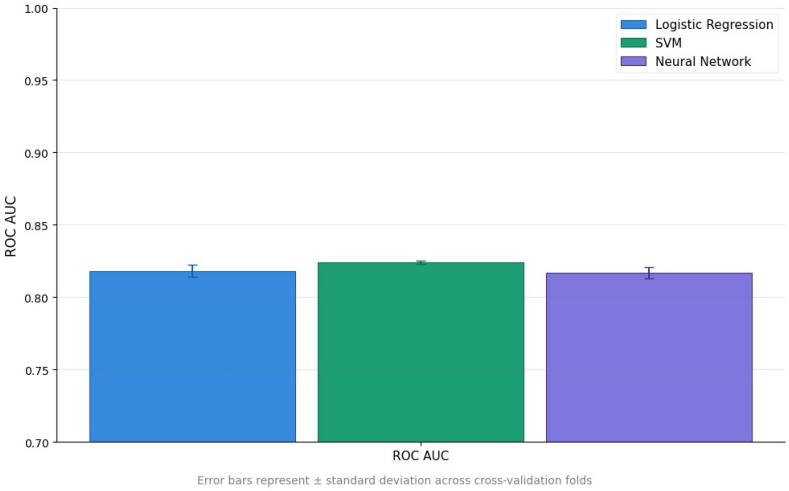

Fig. 2. Comparison of ROC-AUC scores for logistic regression, SVM, and neural networks.

As seen in the graphs, Logistic regression and Neural Networks have the highest variance, however, at 0.004, it is negligible. All models perform quite well on ROC-AUC, and have little to no difference. To better predict which model is best for diabetes classification, a precision/recall curve was used to observe a different evaluation metric for the models, since the ROC-AUC scores were identical. Unlike ROC curves, Precision/Recall curves focus exclusively on the positive class. Model performance was measured using Average Precision (AP), which represents the area under the curve. Finally, a model that is not able to discriminate between classes would achieve an AP equal to the baseline (0.14). As shown in Figure 3, logistic regression outperforms both neural networks and SVM models, with a score of 0.43, approximately three times the baseline prevalence of 0.14. On the other hand, Neural Networks, which scored 0.13, and SVM, which scored 0.14, failed to exceed the baseline. This suggests these models were unable to learn a meaningful decision boundary for this classification task. These results indicate that the logistic regression model is preferred over Neural Networks and SVM. The poor performance of Neural Networks and SVM models may reflect several issues commonly found on imbalanced datasets. SVMs are sensitive to class imbalance, and similarly, neural networks are prone to converging on the majority class. The higher performance of logistic regression in this context suggests that simpler models may outperform more complex ones on imbalanced datasets. However, when considering ROC-AUC as the primary metric, the differences between the models become less pronounced, indicating that their overall ability to distinguish between classes is similar.

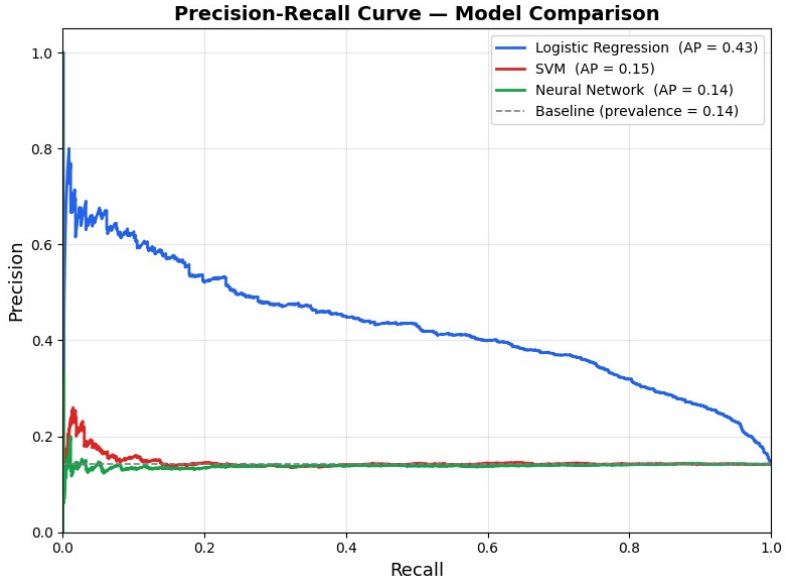

Fig. 3. Precision–Recall curves comparing Logistic Regression, SVM, and Neural Network models on the test set.

5 Conclusion

Overall, all models achieved close ROC-AUC scores, indicating a similar ability to distinguish between diabetic and non-diabetic cases. The similarity in performance can be explained by a few factors. One possible explanation is that simpler models are sufficient to capture the relationship between lifestyle variables and diabetes status. Therefore, increasing the model complexity does not necessarily improve performance in this task.

Even though the models achieved similar ROC-AUC scores, they showed different precision and recall trade-off. In particular, Logistic Regression achieved a higher recall, identifying more diabetics cases, but at the cost of a lower precision, resulting in more false positives. In the context of medical prediction, this trade-off can be acceptable, since incorrectly classifying a non-diabetic individual as diabetic is often less harmful than failing to identify a diabetic case.

The observed differences in recall and precision are related to the underlying characteristics of these models. Logistic regression applies a global decision rule, so after SMOTE balancing it shifts predictions toward the positive class across the entire dataset, resulting in higher recall but lower precision. Meanwhile, SVM focuses primarily on samples near the decision boundary, making it less sensitive to the overall class distribution shift and leading to a more balanced trade-off between recall and precision. Finally, Neural Networks can learn non-linear decision boundaries, which helps them to distinguish between classes and reduce false positives, this results in a higher precision compared to the other models.

As explained previously the ROC-AUC scores are the primary metric of performance in this study. Therefore, the similar ROC-AUC scores suggest that the model choice is not critical for this task. Simpler models such as Logistic Regression achieve similar performance compared to more complex models. These findings help answer the research question by showing that no model outperformed others.

The study had a few limitations such as the highly imbalanced dataset, the exclusion of prediabetic cases, and limited hyperparameter tuning. These factors may have influenced the results. Future work could explore an implementation involving more complex Neural Networks and using non-linear SVM kernels.

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